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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program/s: B Tech

Year: I Semester: II

Stream/s ; Computer Science and Business Systems

Subject: Data Structures and Algorithms

Time: 3 hrs. (2:00pm to 5:00pm)

Date: 25 / 4 / 2024

No. of Pages: 02

Marks: 100

(Re-Examination 2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1_ is compulsory.
2) Out of remaining questions, attempt any _4_ questions.
3) **In all _5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer the following	
CO- 1; BL- L	a.	Explain asymptotic notations used to represent time complexity of an algorithm.	[05]
CO-1; BL- L	b.	Discuss the classification of data structures? Give suitable example.	[05]
CO- 2; BL-M	c.	Differentiate between arrays and linked list.	[05]
CO- 3; BL-M	d.	How do you represent graphs in the memory? Give suitable example	[05]
Q2 CO-2; BL-H	a	Write a program / algorithm to read 10 values in an Array. Perform linear search operation on the array and display the result.	[10]
CO-2; BL-M	b	What is the need of circular queue? Discuss conditions of empty and full queue with suitable example.	[10]
Q3 CO-4; BL-H	a	Apply and explain binary searching algorithm on the following dataset. Search 70 and write all the steps accordingly. 10, 20, 30, 40, 50, 50, 60, 70, 80.	[10]
CO-2; BL-L	b	Discuss any two applications of stack. Perform one push and one pop operation with the help of suitable algorithm on stack.	[10]

Q4 CO-2; BL-M	a	Perform insert and delete operation in between at the specified location of the linked list with suitable algorithms and example.	[10]
CO-3; BL-L	b	Explain all the binary tree traversal algorithms with suitable example.	[10]
Q5 CO-2; BL-M	a	How do you represent a queue using linked list? Write algorithm to perform insert and delete operation on the same.	[10]
CO-3; BL-H	b	Explain the working of Dijkstra Algorithm with suitable example with at least 5 nodes and 8 edges.	[10]
Q6 CO-3; BL-L	a	Explain BFS algorithm with suitable example.	[10]
CO-3; BL-H	b	Given a sequence of elements: 45, 56, 12, 34, 58, 39, 44, 78, 5, 28, 66. Apply and explain the algorithm / function to generate the binary search tree for this given sequence.	[10]
Q7 CO-3; BL-M	a	Discuss the importance of binary tree as expression tree. Draw expression tree for the following infixed expression. 3 * (2 – 4) + (5 * (6 + 7))	[10]
CO-4; BL-L	b	Explain the merge sort algorithm with suitable example	[10]